

SURAJ SAMANTA

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EDUCATION

Dr. Akhilesh Das Gupta Institute of Professional Studies, New Delhi
Bachelor of Technology in Computer Science Engineering | CGPA: 9.4 / 10.0

Aug. 2024 – Jun. 2028 (Expected)

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript (basic)

Backend & Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate, FastAPI, Django, REST APIs, Microservices, Event-Driven Architecture

Distributed Systems: Apache Kafka, gRPC, WebSockets, Scalable System Design

Databases: PostgreSQL, MySQL, TimescaleDB, Redis, Pinecone (Vector DB)

Testing & Build: JUnit, Mockito, Maven, Postman, REST API Testing

DevOps & Tools: Docker, Git, GitHub, CI/CD (GitHub Actions)

Frontend: React.js, Next.js, Three.js / WebGL, Tailwind CSS, Leaflet

ACHIEVEMENTS

Top 5 in India — NASA Space Apps Challenge 2025

Oct. 2025

Competing against 823 teams nationwide in AI-powered geospatial systems

- Built ORBITAIR, an AI-powered air quality forecasting platform using FastAPI and TimescaleDB, integrating NASA TEMPO, EPA/OpenAQ, and NOAA data feeds into a unified pipeline achieving 98% AQI prediction accuracy.
- Developed a React + Leaflet frontend dashboard for real-time geospatial pollution visualization, connected via REST APIs with explainable AI output.

National Finalist — Smart India Hackathon 2025

Dec. 2025

Selected from thousands of national teams to present at the final stage

- Built SAHAI, a full-stack mental health platform with real-time WebSocket chat, RAG-based AI therapy chatbot (Pinecone + FastAPI), and scalable REST APIs across Django and FastAPI services.

Top 5 & ₹7,000 Prize — Smart Delhi Ideathon

Jan 2025

Recognized for civic tech innovation using backend systems and AI

KEY PROJECTS

Patient Management Service | Java, Spring Boot, Kafka, gRPC, Docker, PostgreSQL [Suraj-codes1410/Patient-management-services](https://github.com/Suraj-codes1410/Patient-management-services)

- Engineered 4 independently deployable microservices in Spring Boot, reducing inter-service coupling and enabling isolated scaling of billing, auth, and patient data modules.
- Cut inter-service latency by implementing gRPC for synchronous RPC alongside Apache Kafka for async event streaming, supporting high-throughput pipelines between services.

SAHAI — Mental Health & Lifestyle Platform | Django, FastAPI, React, WebSockets, MySQL, Pinecone
Smart India Hackathon 2025 — National Level Finalist Nov. 2025 – Jan. 2026

- Architected a full-stack mental health platform with appointment booking, live chat, and AI therapy features designed for 1,000+ concurrent sessions via a normalized MySQL relational schema.
- Built a RAG microservice with Pinecone vector DB to improve AI chatbot response relevance by indexing structured therapy content, reducing irrelevant responses vs. a base LLM.

Multiverse OS — Immersive Developer Portfolio | Next.js, React, Three.js, WebGL, Zustand, Tailwind CSS, Vercel
[Suraj-codes1410/Multiverse-OS](https://github.com/Suraj-codes1410/Multiverse-OS)

Deployment in progress — fully built, CI/CD pipeline configured

- Built a full-stack interactive portfolio as a simulated OS in Next.js and React, featuring a 3D WebGL desktop rendered with Three.js and animated scene transitions via custom WebGL shaders.
- Engineered a CLI terminal overlay with 15+ live commands, an ORACLE AI assistant with context-variant prompts, and a Recruiter Mode dashboard — all state managed via Zustand.
- Designed the full 6-phase system architecture — component registry, API layer, animation system — with CI/CD automated deployment to Vercel via GitHub Actions.